

RoomRadar QC — Software Limitations

Project: RoomRadar QC

Course: CSCI 370

Date of Delivery: May 2026

1. Data Is Not Real-Time

RoomRadar QC displays room availability based on the semester schedule uploaded by an admin. It does not reflect last-minute cancellations, ad-hoc reservations, or any changes made to the schedule after the CSV was uploaded. A room shown as available may still be in use if the schedule changed.

2. Manual CSV Upload Required

Room data must be manually uploaded by an authenticated admin using the Queens College schedule CSV format. There is no automated sync with any official Queens College scheduling system. If the schedule is updated mid-semester, an admin must re-upload the new CSV to reflect those changes.

3. Queens College Specific

The system is built specifically for Queens College. Building codes (PH, KY, SB, RE) and the CSV format are hardcoded for Queens College's scheduling data. The application cannot be used for other campuses or institutions without code changes.

4. No Room Booking or Reservations

RoomRadar QC only shows which rooms appear to be unscheduled during a given time window. It does not allow users to reserve or book a room. There is no guarantee that a room displayed as available will remain available by the time the user arrives.

5. No Room Feature Filtering

Users cannot filter rooms by features such as projector availability, computer lab setup, seating capacity, or accessibility. The system only filters by building and time window.

6. Single Active Semester

The room search does not filter by semester. All schedules across all uploaded semesters are considered when determining availability. If multiple semesters are uploaded, older semester data may affect search results.

7. CSV Format Dependency

The admin upload only works with the specific CSV format exported from Queens College's scheduling system. CSVs with missing columns, different column names, or different time formats will cause the upload to fail or produce incorrect data.

8. No Offline Support

The application requires an active internet connection to function. The frontend is hosted on Vercel and the backend on Render, both of which require connectivity. There is no offline or cached mode.

9. Render Cold Starts

The backend is hosted on Render's free tier, which spins down the server after a period of inactivity. The first request after inactivity may take 30–60 seconds to respond while the server restarts.

10. No Multi-Factor Authentication for Admin

Admin access is protected by Firebase email/password authentication only. There is no multi-factor authentication, meaning a compromised admin account could allow unauthorized CSV uploads or data manipulation.